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RESEARCH ARTICLE

Adaptive Canny and Semantic Segmentation Networks Based on Feature Fusion for Road Crack Detection

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ABSTRACT Road cracks, which are a common hazard in pavements throughout the life cycle of a road, can degrade the performance of the road, shorten its service life, and endanger the safety of vehicles. Traditional vision machine detection methods can detect road crack details but suffer from poor stability and generalization ability, whereas semantic segmentation detection, although more stable, cannot track fine road crack information. To combine the advantages of both methods and improve the accuracy of road crack detection, a novel feature fusion road crack detection method is proposed in this study. First, the bilateral filter and four-way Sobel operator are introduced into the Canny algorithm to enhance the noise reduction effect and extract edge features more effectively. Second, the dynamic threshold is generated adaptively by the gradient information after non-maximum suppression. Subsequently, the detection map is morphologically processed, the connected areas are ranked, and the bilateral filter parameters are adjusted based on the detection results. The Canny road crack detection map is then extracted by the convolutional feature extraction module, fused with the low feature layer in the DeepLabV3+ detection network, and finally stitched with the high feature layer; the resulting map is obtained after convolutional feature extraction. The method was validated on the publicly available complex road crack dataset CRACK500; the experimental results showed that feature fusion outperformed the adaptive Canny, DeepLabV3+, Unet, PSPnet, and ICnet algorithms by more than 6.5% on the Mean Intersection over Union(MIoU) and also in Mean Absolute Error(MAE) by effectively combining crack features and improving the detection accuracy.

INDEX TERMS Adaptive canny, feature fusion, road cracks, semantic segmentation.

I. INTRODUCTION

Cracks are one of the most common road defects. The appearance of cracks indicates that the road structure has been damaged to varying degrees, allowing corrosion to reach the internal reinforcement layer of the road, whereby internal damage to the road is accelerated [1], and the tensile strength of the road is lowered; this not only affects the aesthetics of the road but also poses a potential threat to pedestrian safety [2], [3]. Therefore, road crack detection must be performed regularly on roads. Previously, road crack detection was performed manually; however, manual inspection was time-consuming and laborious; the results were also susceptible to the subjective awareness of different inspectors. The subsequent rapid development of computer vision detection algorithms led to them gradually being applied; however, the influence of several interfering factors, such as complex road materials, excessive pollutants, and weather changes, increase the probability of false detection and missed detection errors in the inspection process.

With the development of computer vision and artificial intelligence, the use of machines for road crack detection has become increasingly common. The use of computers renders the road crack detection process less susceptible to complex

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and changing environments, and the detection results are more standardized and accurate. Currently, two mainstream machine detection methods are road crack detection based on traditional vision and road crack detection based on semantic segmentation.

In the traditional visual inspection method, image processing technology is employed to preprocess images and reduce noise interference. Subsequently, a featured operator is used to extract the characteristic information of the road crack, enabling differentiation between the cracked and non-cracked regions. However, traditional image processing suffers from several limitations, including the need to set multiple parameters, lack of self-learning ability, vulnerability to environmental noise, poor adaptability to changes in the detection environment, and limited generalization ability. In the semantic segmentation detection method, a convolutional neural network is trained to identify road crack features and classify each image pixel. While this approach is more stable, some issues arise in actual road crack scenarios, such as pixel ratio disparities between the road crack and background. As a result, the semantic segmentation detection method may overlook crucial details in the road crack.

To address the issue of excessive manual parameter adjustment, poor stability in traditional visual inspection methods, and the loss of detail associated with semantic segmentation detection, this paper proposes a road crack detection method based on feature fusion, adaptive Canny, and DeepLabv3+. In traditional image processing, the four-directional Sobel operator is used instead of the original two-directional Sobel operator. Bilateral filtering replaces the original Gaussian filtering to enhance feature extraction and noise reduction. Adaptive bilateral filtering is obtained through feedback from the results, and adaptive Canny thresholds are obtained from the image gradient information, improving the automation level of detecting complex road cracks. The adaptive Canny detection image is fused with the low-level features of the DeepLabV3+ network through a feature extraction module to solve the problem of losing some details caused by the combination of high and low-level features, thereby improving the accuracy of road crack detection.

This paper selects the Canny edge detection operator as the traditional image processing method. There are many edge detection operators, and Wang et al. [4] compare five commonly used operators and come to the following conclusions: the Roberts operator has good localization but detects fewer edge details; the edge detection effect of Sobel is similar to Prewitt, with clear contours, but wider edges; the Kirsch operator detects blurry edges and has low localization accuracy. The Canny operator uses a Gaussian function to smooth the image before detecting the edges. It is not easily disturbed by noise and can detect true weak edges. Two different thresholds are used to detect strong and weak edges. When weak edges are connected to strong edges, the weak edges are included in the output image. The detected edges are more complete, better connected, and have higher localization accuracy. Therefore, the Canny operator is chosen as the traditional image processing method for its excellent edge detection ability and suitability for crack detection. This paper selects DeepLabv3+ as the semantic segmentation network. DeepLabv3+ [5] is an extension of DeepLabv3 and is considered a new pinnacle in semantic segmentation by the academic community. DeepLabv3+ combines the advantages of the spatial pyramid pooling model and the encoder-decoder structure. It adds a simple and effective decoder module to refine the segmentation results, especially along the object boundaries. The authors further explore the application of depth-separable convolution to the atrous spatial pyramid pooling and decoder module, resulting in a faster and stronger encoder-decoder network. The DeepLabv3+ network has an advanced architecture, excellent segmentation accuracy, and enormous potential to handle surface feature segmentation tasks in crack scenarios [6]. Therefore, DeepLabv3+ is chosen as the semantic segmentation part, and many researchers use DeepLabv3+ for crack detection, which will be discussed later in this paper.

In this study, the primary contributions presented in this article are as follows:

- An adaptive Canny dual-threshold selection method is proposed in this paper. The method generates a gradient histogram by traversing the image pixels. It calculates the gradient region most likely to be the road crack edge based on the gradient variance and sets high and low thresholds accordingly. Experimental results show that the adaptive Canny threshold set according to the gradient histogram can effectively detect road crack edges and solve the problems of inaccurate and inefficient manual threshold setting.
- An adaptive bilateral filtering method is proposed in this paper. The method adjusts the filtering parameters through feedback from the detection results. Firstly, the number and area of connected regions of road crack edge images generated are detected. If there are multiple small connected regions, it indicates that the noise interference is too large and the filtering strength needs to be increased. If the connected regions are few and small, it indicates that some road crack information is lost, and the filtering strength needs to be reduced.
- A supervised learning method for feature fusion of adaptive Canny detection results and DeepLabV3+ network is proposed in this paper. The Canny result image is fused with the low-level feature layer in the DeepLabV3+ network, increasing the road crack position and details and compensating for the loss of details when the high and low-level feature layers are fused. These details can detect small cracks. The proposed fusion method is tested on the literature's widely used CRACK500 complex road crack dataset. The experimental results show that this method is superior to the prefusion method in terms of MIoU and MAE values and has better results than previous semantic segmentation networks.

In the following sections, Section II introduces related research on road crack detection; Section III presents the

method proposed in this study; Section IV presents the implementation details, experimental results, and analysis; and Section V provides the conclusions.

II. RELATED WORK

This section introduces research on road crack detection and their advantages and disadvantages. Road crack detection based on computer vision includes traditional visual and semantic segmentation detection.

A. ROAD CRACK DETECTION BASED ON TRADITIONAL VISION

Following the traditional vision for road crack detection. Huan et al. [7] proposed an improved Canny operator for pavement road crack detection. They used morphological filtering to replace the original Gaussian filtering and Otsu's maximum inter-class variance threshold segmentation algorithm to achieve adaptive determination of dual thresholds. The experimental results show that the improved algorithm demonstrates improved crack detection speed and accuracy, but only for relatively simple crack scenes with little noise interference, and performs poorly in more complex crack environments. Zhao et al. [8] further improved the Canny operator and applied it to crack detection. They replaced the original Gaussian filter with bilateral filtering. They used the Otsu algorithm to adaptively obtain a double threshold and morphological filtering to eliminate small voids and fill cracks in the contour lines. The improved method enhances the accuracy of detection but is still ineffective in detecting cracks in complex environments and has weak generalization capability; it is thus unable to meet the needs of simultaneous multi-scene crack detection. Monicka et al. [9] proposed an improved Canny operator for noise removal that incorporates adaptive threshold selection, morphological operations, image segmentation, and image binarization. The experimental results show that the improved algorithm is more accurate than traditional edge detection algorithms in detecting cracks. However, the problem of too many manually adjusted parameters persists, which affects detection efficiency. To improve the Canny operator, Meiling et al. [10] first used statistical filtering to solve the image edge blurring problem, thereby calculating the gradient amplitude and direction of the crack image based on the 3×3 gradient template of the operator. Finally, the natural iteration method was used to obtain the image threshold, and discontinuous edges were repaired using the morphological closure operation and thinning operators to obtain the final detected edges. Experiments showed that this improved algorithm maintains the integrity of the detected edges and can reduce the interference of granular and flaky noise. Akagic et al. [11] proposed using Otsu-thresholded image segmentation for pavement crack detection. First, the method divides the input image into four independent sub-images of equal sizes. Then, cracks are searched based on the ratio between the Otsu threshold and the maximum histogram value of each sub-image, and all segmented sub-images are merged. This method performs better with low signal-to-noise ratios but is unsuitable for complex crack environments.

B. ROAD CRACK DETECTION BASED ON SEMANTIC SEGMENTATION

Semantic segmentation, in this study's context, involves classifying each image pixel for road crack detection. Zhang et al. [12] developed CrackNet based on a convolutional neural network to detect cracks at the pixel level; CrackNet does not have any pooling layers, and its feature map size is constant for all layers. Subsequently, in 2018, Zhang et al. [13] proposed the improved CrackNet II. Compared with CrackNet, CrackNet II sets trainable parameters in all hidden layers to further improve crack detection performance. Fei et al. [14] proposed CrackNet-V on top of CrackNet, which significantly improves the crack detection rate. The above series of CrackNet models can maintain spatial resolution and avoid the loss of spatial information during downsampling. In addition, encoder-decoder architectures have been widely used for semantic segmentation. Among the various encoder-decoder architectures for road crack detection, DeepLabv3+ is a widely used technique. Using their dataset, Ji et al. [15] were the first to apply DeepLabv3+ to road crack detection. The effectiveness of DeepLabv3+ was verified for single, multiple, intersecting, and crocodile cracks, and its performance was superior to that of FCN [16] and DeepCrack [17]. However, DeepLabv3+ suffers from poor performance in crack details, which could be attributed to the suppression of some crucial information in the high feature layer by the low feature layer during fusion [18]. Attention mechanisms have been used in some studies to address this issue. Sun et al. [19] proposed a multi-scale attention module in the decoder of DeepLabv3+ to generate attention masks and dynamically allocate weights between high-level and low-level feature maps. Dynamic weights can assign more rational weights to different feature maps than fixed weights across different features. The network exhibited significantly improved segmentation performance and performed well on the road crack dataset. Liu et al. [20] improved DeepLabv3+ by using Resnet50 as the backbone network to enhance its feature extraction capability. While this improved method showed higher detection accuracy than other common pixel-level segmentation algorithms, it required increased time due to the increased network layers. Cai et al. [21] utilized MobileNetV2 as the backbone feature extraction network, substantially reducing the model's computational load and increasing the calculation speed. Furthermore, they introduced the attention mechanism module into the backbone feature extraction network and decoder, further optimizing the model's edge recognition effect and segmentation accuracy.

TABLE 1. Summary table of work related to road crack detection.

Thesis Tittle	Research Methods	Advantage	Disadvantages
[7]	Adaptive Selection of Canny Double Thresholds Using Ostu Algorithm	Avoid the tedious manual parameter adjustment	Does not solve the interference of road environmental noise
[8]	Use the Ostu algorithm to obtain the threshold and introduce morphological filtering	Improved detection accuracy	Susceptible to environmental interference, poor detection effect on complex cracks
[9]	Added adaptive threshold selection, morphological operations, image segmentation	Better ability to remove noise	Many parameters need to be set manually
[10]	Solving Blurred Edges with Statistical Filtering	Preserves edge integrity and reduces grain and flaky noise interference	
[11]	Use Otus to segment images, detect them and merge them	Better performance at low signal-to-noise ratios	Not suitable for complex crack environment
[12]	Semantic segmentation network developed based on CNN	Maintain spatial resolution to avoid loss of spatial information during downsampling	The detection speed is not fast and the accuracy is not high
[13]	Trainable parameters are set in all hidden layers	Improved performance for detecting cracks	
[14]	Compared to the original version of CrackNet, CrackNet-V has a deeper architecture but fewer parameters	Improved detection accuracy and computational efficiency	
[15]	A Semantic Segmentation Network Based on Encoder-Decoder Architecture	Clear boundary extraction	Loss of crack details during feature layer fusion
[19]	Introducing multi-scale attention modules into the decoder	Enhance the ability to detect details	Insufficient generalization ability
[20]	Select Resnet50 as the backbone network for improving DeepLabv3+	Enhanced feature extraction capability	Slow down operation speed
[21]	Using MobileNetV2 as the backbone feature extraction network and introducing attention mechanism	Enhance edge extraction capability	

Table 1 provides a summary of related work. The road crack detection method based on traditional vision can detect the edges of road cracks and even minor details in environments with sufficient lighting and minimal noise interference. Its detection accuracy is superior to that of semantic segmentation. However, this approach is highly susceptible to environmental factors such as pollutants, lighting, and road signs, which can lead to inaccurate detection results. Moreover, the manual parameter setting is laborious, and the method has weak generalization ability and low efficiency. On the other hand, the road crack detection method based on semantic segmentation is less prone to environmental disturbances, and it exhibits strong generalization ability and high robustness. However, it needs improvement to detect crack details effectively.

III. METHODS

This section presents the proposed method in this paper, which comprises three parts: the adaptive Canny method, the DeepLabv3+ method, and the fusion of the two.

This study proposes feature fusion-based adaptive Canny and DeepLabV3+ for crack detection, which is used for pixel-level crack detection. The flowchart is shown in Fig. 1. The proposed method is divided into two parts: (1) adaptive Canny for crack detection in the original image and (2) adaptive Canny detection with DeepLabV3+ network fusion for crack detection in the resulting image. The original incoming image is preprocessed in the first part to reduce ambient noise interference by grayscaling and bilateral filtering. Second, the gradient operator is used to calculate the gradient magnitude and direction of the pixel points of the entire image, whereby a suitable threshold is generated based on the gradient map to complete edge detection. Finally, morphological operations are used to refine the edge-detected image. The original image is fed into the DeepLabV3+ network for detection in the second part, divided into two paths after the dynamic convolutional neural network (DCNN). One path consists of the low feature layer, which is to be fused with the detection result map of the first part after the convolutional feature extraction module; the other path is passed through the enhanced feature extraction network known as the atrous spatial pyramid pooling network for information extraction to generate the high feature layer. Subsequently, the high and low feature layers are stacked, followed by two depth-separable convolution blocks, and then upsampled to the size of the input image to complete the final prediction.

A. ADAPTIVE CANNY DETECTION

1) IMAGE PRE-PROCESSING

The publicly available complex road dataset CRACK500 covers a wide range of cracks caused by factors such as road material and contaminants. To address the complexity of the pavement images, the crack images must be preprocessed before they are fed into the inspection system, as outlined in the following steps:

(1) Grayscale. The acquired dataset images are in color. To reduce memory consumption and increase the processing speed of the images, they must be converted into 8-bit greyscale images to reduce the amount of data for processing road images [22].

(2) Bilateral filtering [23]. The grayscale image still contains several environmental distractors affecting the subsequent edge feature extraction; therefore, filtering is required. The original Canny algorithm applies Gaussian filtering to the image. Gaussian filtering is a weighted average of neighboring pixels in the domain and can be used to remove high-frequency noise. However, the edges are also high-frequency signals, and some edge information of the cracks is lost while eliminating interference noise, which affects the subsequent extraction of fine crack features. This study used bilateral filtering to replace the original Gaussian filter.

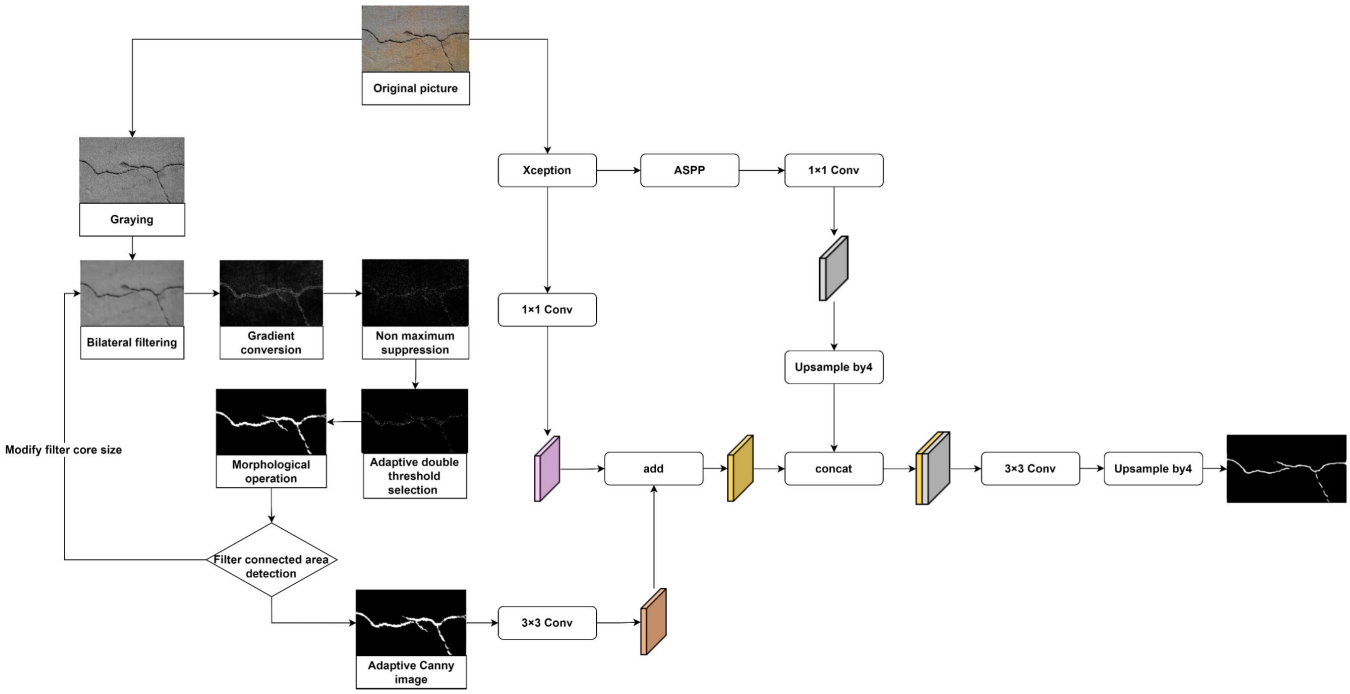

FIGURE 1. The block diagram of the crack detection method proposed in this paper is presented below. The left half of the diagram shows the adaptive Canny detection process, followed by fusing the resulting image with the DeepLabV3+ network structure in the right half for further detection.

Bilateral filtering is a nonlinear filter that can maintain edges and reduce noise smoothly and is thus ideal for crack pictures of complex pavements. Bilateral filtering uses a weighted average method [24], in which the intensity of a pixel is represented by the weighted average of the luminance values of the surrounding pixels; this weighted average is based on a Gaussian distribution. Most importantly, the weights of the bilateral filter consider not only the Euclidean distance of the pixel but also the radiometric differences in the pixel range domain, including the degree of similarity between the pixel in the convolution kernel and the central pixel, color intensity, and depth distance, among others. Both weights are considered when calculating the central pixel, and the formulae are shown in (1) and (2).

$$I_p^{bf} = \frac{1}{W_p^{bf}} \sum_{q \in s} G_{\delta_s}(p-q)\, G_{\delta_r}\left(\left|I_p - I_q\right|\right) I_q \tag{1}$$

$$W_p^{bf} = \sum_{q \in s} G_{\delta_s}(p-q)\, G_{\delta_r}\left(\left|I_p - I_q\right|\right) \tag{2}$$

where: I_p is the input image and I_q is the output image; p and q are any two points on the image; s is the range region centered on a point; I_p^{bf} is the filtered image; W_p^{bf} is the normalization factor; δ_s is the spatial domain standard deviation; δ_r is the value domain standard deviation; $G_{\delta_s}(p-q)$ is the spatial weight and $G_{\delta_r}\left(\left|I_p - I_q\right|\right)$ is the pixel range weights. For the points to be filtered, the weight of the pixel point in its domain is related to the distance and similarity between the two pixel points. Combining the spatial distance and degree of similarity, two-sided filtering can be achieved using the formulae shown in (3) and (4).

$$h(x) = k^{-1}(x) \int_{+\infty}^{-\infty} \int_{+\infty}^{-\infty} f(\varepsilon)\, c(\varepsilon, x)\, s \times [f(\varepsilon), f(x)]\, d\varepsilon \tag{3}$$

$$k(x) = \int_{+\infty}^{-\infty} \int_{+\infty}^{-\infty} c(\varepsilon, x)\, s\, [f(\varepsilon), f(x)]\, d\varepsilon \tag{4}$$

where: $h(x)$ is the filtered grey value of point x; c is the weight of Gaussian space; s is the Gaussian weight of the degree of similarity between pixels [25]; $f(x)$ is the pixel value of the pixel point; $k^{-1}(x)$ is the normalization factor.

To obtain a better result map, the parameters of c, s and filter kernel size must be set manually. Manual adjustment is inefficient because the dataset contains many images of different types. To address this problem, this study proposes an adaptive bilateral filtering method, which is presented in the subsequent sections.

2) CALCULATING IMAGE GRADIENTS

The Canny algorithm uses the traditional Sobel operator, which only considers gradient information in both directions, is not robust in analyzing images, and is easily affected by noise. It does not consider the image edge direction and often loses edge details [26].

To reduce the influence of noise on the image, enhance the detection of contour lines, and retain edge details, this study drew on the first-order gradient template of the Sobel

operator to determine the gradient amplitude of the image, thereby extending it to a first-order gradient template in four directions: horizontal, vertical, 45°, and 135° [27], as shown in Fig. 2.

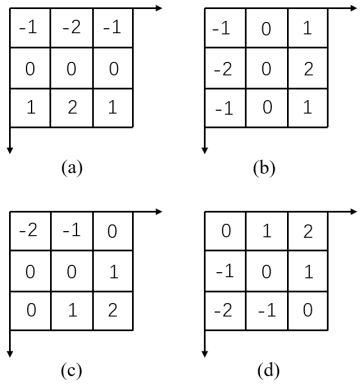

FIGURE 2. Gradient direction template. Where a is the x-axis direction, b is the y-axis direction, c is the 45°-axis direction, and d is the 135°-axis direction.

The first-order gradient components $G_x(x, y)$, $G_y(x, y)$, $G_{45^\circ}(x, y)$ and $G_{135^\circ}(x, y)$ in each of the four directions can be obtained by convolving the filtered image with each of the four first-order gradient templates in Fig. 2.

The gradient amplitude and angle can be obtained from the first-order gradient components in the four directions, as shown in (5) and (6).

$$M(x, y) = \sqrt{G_x^2 + G_y^2 + G_{45}^2 + G_{135}^2} \tag{5}$$

$$\theta(x, y) = \arctan\left(\frac{G_y}{G_x}\right) \tag{6}$$

3) CANNY ADAPTIVE THRESHOLD SELECTION

In response to the problem of the traditional Canny algorithm requiring manual adjustment of high and low thresholds, this study proposes an adaptive dynamic thresholding method. Gradient conversion is performed on the image after non-maximum suppression, converting the gradient value to 0 and 255. The gradient magnitude of each pixel point is counted to generate a gradient histogram describing the edge strength [28]. The gradient histogram has information on image edge strength, where pixel points with gradient 0 indicate non-edge regions that are excluded to not affect the subsequent calculation. The remaining points contain all the detected edge information, which contains both correct and incorrect edges, that is, ambient noise, as shown in Fig. 3.

The gradient histogram demonstrates no obvious bimodal peak, only a very high single peak in the region of low gradient values, indicating that the ambient noise in the image accounts for the majority of the cracked image. The region corresponding to the edges in the original image in the gradient histogram shows no obvious spikes but exhibits a uniformly decreasing state with an approximately normal distribution. There is no obvious valley boundary between the non-edge and edge regions in the gradient histogram;

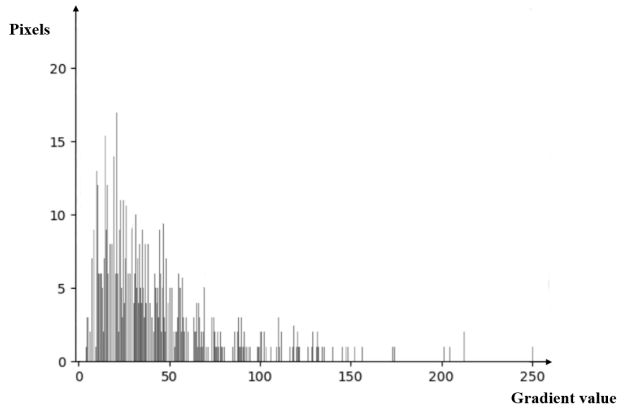

FIGURE 3. Gradient histogram of the crack image.

therefore, the gradient histogram cannot be segmented by simply using the grayscale histogram to determine the bimodal peak.

Based on the properties of the gradient histogram, a method is proposed for adaptively segmenting segment the correct edge region from the incorrect edge region. The images of the road crack dataset contain both crack and environmental information, and it is known, from a priori knowledge, it is known that the proportion of incorrect edges in the images is much larger than the proportion of correct edges; therefore, the pixel-level sum of the areas corresponding to the peaks of single peaks and their vicinity in the gradient histogram must be the set of incorrect edges, that is, environmental interference noise. The gradient with the highest number of pixels in the gradient histogram is called the pixel-maximum gradient G_{max}, and the variance of the gradient of all pixels within the sub-image relative to G_{max} is calculated and referred to as the pixel-max gradient variance σ_{max}.

$$\sigma_{max} = \sqrt{\sum_{i=1}^{255} (i \times G_i - max \times G_{max})^2 / \sum_{i=1}^{255} (i \times G_i)} \tag{7}$$

where i is the gradient value, and max is the gradient value containing the maximum number of pixels.

When there is little ambient noise, the gradient histogram has only a single peak, the pixel gradient values are concentrated around the most-valued gradient G_{max}, and σ_{max} is small. When the gradient histogram reflects not only cracked edges but also a large proportion of false edges caused by ambient noise, the correct edge pixel gradients are distributed relatively far from G_{max}; consequently, σ_{max} is large. Therefore, based on this analysis, high and low thresholds can be adaptively set.

Analysis of the principle of the Canny algorithm shows that the double-threshold connection ignores weak edge points below the low threshold τ_l and retains strong edge points above the high threshold τ_h. Edge points between τ_l and τ_h are retained depending on whether there are strong edge points in the neighborhood. Therefore, τ_l must be set outside the wrong edge region. Otherwise, a large amount of false edge

noise will be introduced into the final result. In this study, we set τ_l adaptively. G_{max} reflects the central position of the distribution of the error edge, that is, ambient noise, in the gradient histogram, whereas σ_{max} reflects the dispersion of the gradient distribution in the gradient histogram relative to G_{max} that is, the dispersion relative to the error edge. Therefore, it can be used as G_{max}, and the σ_{max} is used to calculate the extent of the erroneous edge region.

The distribution pattern of each gradient pixel in the gradient histogram conforms to a normal distribution, and it is empirically known that the gradient values of the crack edges are larger so that few correct edge regions are in regions with larger gradient values. According to the 3σ principle for the normal distribution, when the low threshold value τ_l is greater than the pixel-maximum gradient G_{max} by three times the pixel-maximum gradient variance σ_{max}, τ_l, is considered outside the wrong edge region; thus the environmental noise edge can be prevented from appearing in the contour map. The formula for calculating τ_l is,

$$\tau_l = G_{max} + 3 \times \sigma_{max} \tag{8}$$

Once τ_l has been determined, τ_h is generally considered to be three times the low threshold τ_l.

4) MORPHOLOGICAL MANIPULATION AND ADAPTIVE BILATERAL FILTERING

The morphological closure operation [29] is image expansion followed by erosion. When applied to crack images, this process removes narrow interruptions and long thin divots, eliminates small holes, and replenishes cracks in the contour lines. This operation smoothens the contour edges.

After the morphological closure operation, the edges demonstrate superior smoothing and noise reduction while retaining more image edge details. At this point, only the crack edge information is obtained, and the center of the crack is an undetected area, forming a large hole that must be eliminated by morphological reconstruction of the image. After morphological reconstruction, crack information is fully detected, but some environmental noise is also detected. The non-cracked areas are filtered out by determining the connected areas and comparing the size of each connected area across the image.

As mentioned earlier, the parameters c, s, and the filter kernel size must be set manually to obtain a better bilateral filtering effect. This study proposes an adaptive method with feedback adjustment based on the detection results to solve this problem.

(1) The size of the filter kernel. Filter kernel size determines the number of pixels involved in filtering calculations. A larger filter kernel leads to better denoising but requires greater computational effort and generates a blurred image. The filtering speed is increased to reduce the amount of computation; to maintain a good filtering effect, the kernel size is generally chosen to be no larger than 15, depending on the situation.

(2) Gaussian spatial weight. A linear relationship is observed between the Gaussian spatial weight and the size of the filter kernel [30]. Because more than 95% of the components of the Gaussian function are concentrated in the interval $[-2\sigma, 2\sigma]$, to obtain a larger number of samples while ensuring image clarity, the spatial weight is defined as,

$$c = \frac{mr}{2} \tag{9}$$

where c is Gaussian spatial weight, m is a constant, and r is the filter kernel size. The best value of m is experimentally determined to be in the range of [0.75-0.85], which can effectively over-disperse the spatial weight, resulting in blurred images.

(3) Gaussian pixel weight. Compared with Gaussian spatial weight, the effect of bilateral filtering is more easily influenced by Gaussian pixel weight [31]. The feedback from the detection result map is used to adjust Gaussian pixel weight, which can be more influential in bilateral filtering. When the percentage of the maximum connected area is too large, and the number of connected areas is too many, it indicates that noise significantly impacts the detection area and that Gaussian pixel weight should be increased. Gaussian pixel weight is adjusted according to two feedback indicators: the maximum connected area ratio and the number of connected areas. After conducting numerous experiments, it was concluded that the detection region of a correct crack should be the largest connected region in the detection map, comprising 12% to 18% of the entire image area, and the number of connected regions should not exceed 5. In the adaptive bilateral filtering method proposed in this paper, the filter kernel size was set to 15, the Gaussian spatial weight was set to 8 using formula (9), and the initial value of the Gaussian pixel weight was set to 15. Next, the image was predicted, and the detection results were looped through. If the largest connected area accounted for more than 18% of the entire image area, or the number of connected areas was greater than 5, it indicated that the image was significantly disturbed by noise and that the algorithm had misidentified some non-crack areas. Gaussian pixel weight should be increased to improve the filtering effect. Thus, the value of s was incremented by 1, and the detection was re-filtered. Conversely, if the largest connected region accounted for less than 12% of the entire image area, or the number of connected regions was less than 5, it indicated that no cracks or some cracks were missing. In this case, Gaussian pixel weight should be reduced to weaken the filtering effect. Therefore, the value of Gaussian pixel weight was decremented by 1, and the detection was re-filtered.

B. SEMANTIC SEGMENTATION AND FEATURE FUSION

1) DeepLabV3+

DeepLabV3+ is an excellent semantic segmentation model that introduces the encoder-decoder form to fuse multiscale information better. The encoder-decoder architecture introduces arbitrary control over the resolution of the features extracted by the encoder and balances accuracy and time

consumption through atrous convolution. The DeepLabV3+ model is divided into two parts: one for the encoder and the other for the decoder.

The encoder uses serial atrous convolution in the backbone network DCNN. After applying deep convolutional feature extraction to the incoming image, the output is divided into two parts: one part is directly passed to the decoder as the lower feature layer, and the other part goes through parallel atrous convolution at different rates to extract features, capture contextual information at different scales, merge them, and perform 1×1 convolution to compress the features to obtain the high-level feature layer for the decoder.

In the decoder, the higher feature layer from the encoder is upsampled by bilinear interpolation using a magnification factor of four; the concat feature is then used for fusion with the lower feature layer after 1×1 convolution to compress the number of channels. Two depth-separable convolution blocks are then used to extract features, which are then upsampled and magnified four times to obtain a prediction map of the same size as the original image.

2) FEATURE LAYER FUSION

In image processing, feature fusion at different scales is an important tool for improving image results. In convolution, lower-level features have higher resolution and contain more location and detail information but are less semantic and noisier. In contrast, higher-level features have stronger semantic information at the expense of very low resolution and poor detail perception. Therefore, fusing features from different scales is the key to improving the results. In the DeepLabV3+ model, the high and low feature layers are stitched together using concat [32], and the number of features describing the image (the number of channels) is increased; however, the information under each feature does not increase, and the low feature layers possess less semantic information and play little role in the final prediction [33]. When combining different levels of feature layers, the misalignment of information feature locations may lead to a loss of detail in detecting small cracks [18]. In this study, we propose fusing the adaptive Canny detection map with lower feature layers after feature extraction and performing sequential operations. The structure of this method is shown in Fig. 4.

This paper proposes a feature fusion method that fuses the adaptive Canny result map with the low feature layer of DeepLabv3+. This approach was chosen for three main reasons. Firstly, the Canny detection map is rich in detail and provides accurate position information. The low feature layer has finer features than the high feature layer, which enhances its ability to locate object positions and borders. Hence, selecting and fusing low-feature layers for pixel-level feature fusion can improve the accuracy of semantic segmentation. Secondly, low-feature layers are more sensitive to pixel-level details, such as texture and edge, which must be considered when fusing pixel-level features. As a result, selecting and fusing with low-feature layers can better preserve these details, improving the accuracy of semantic segmentation. Finally, compared to fusing Canny detection maps with high-feature layers, fusing with low-feature layers requires fewer convolution kernels, making the algorithm run faster.

In the structure diagram, after adaptive Canny detection, the original map becomes a Canny detection map, to which a 3×3 convolutional block is applied for feature extraction to obtain a low feature layer containing more semantic information. This feature layer is then fused with the lower feature layer after DCNN, with an additional fusion considered at this point.

Compared to concat, instead of increasing the number of features (number of channels) of the image, the dimension of each feature layer needs to be superimposed directly so that the amount of information under the features describing the image is increased, which is equivalent to addition in advance and an increase in the amount of semantic information in the otherwise low-feature layers, which is beneficial for the final classification.

IV. EXPERIMENTAL DETAILS

This section presents the experimental section of the method proposed in this paper. It includes the description of the experimental environment, the parameter settings, the selection of the dataset, the ablation experiments, the comparative experiments with other semantic segmentation networks, and the experimental analysis.

In the experimental study, the CRACK500 dataset and GAPS384 dataset [34] were used to evaluate the performance of the proposed method. The adaptive Canny, DeepLabV3+, Unet, PSPNet, and ICnet methods for pixel-level surface defect detection were evaluated, and their performances were compared with the proposed method.

The hardware environment used to conduct the experiments in this study was an Intel Xeon(R) W-2145 CPU with an NVIDIA GeForce RTX 2080Ti GPU with 11 GB of graphics memory; the software environment consisted of Ubuntu 18.04, Python 3.6, PyTorch 1.10, and the associated Python libraries for neural networks.

Information on the parameter settings of the model as well as numerical and visualization results, are provided.

A. PARAMETER SETTINGS

This study developed the proposed method using Anaconda, Python, and PyTorch. A supervised learning method is utilized to train the network model, and the corresponding details and parameter selections are provided below.

- Bilateral filter kernel size of 15, Gaussian spatial weight c of 8, and Gaussian weight s of 15 to determine the degree of similarity between pixels.
- Minimum connected pixel area of 1500,
- Batchsize of 12,
- Initial learning rate of 0.001 and minimum learning rate of 0.00001,
- Number of training iterations was set to 100,

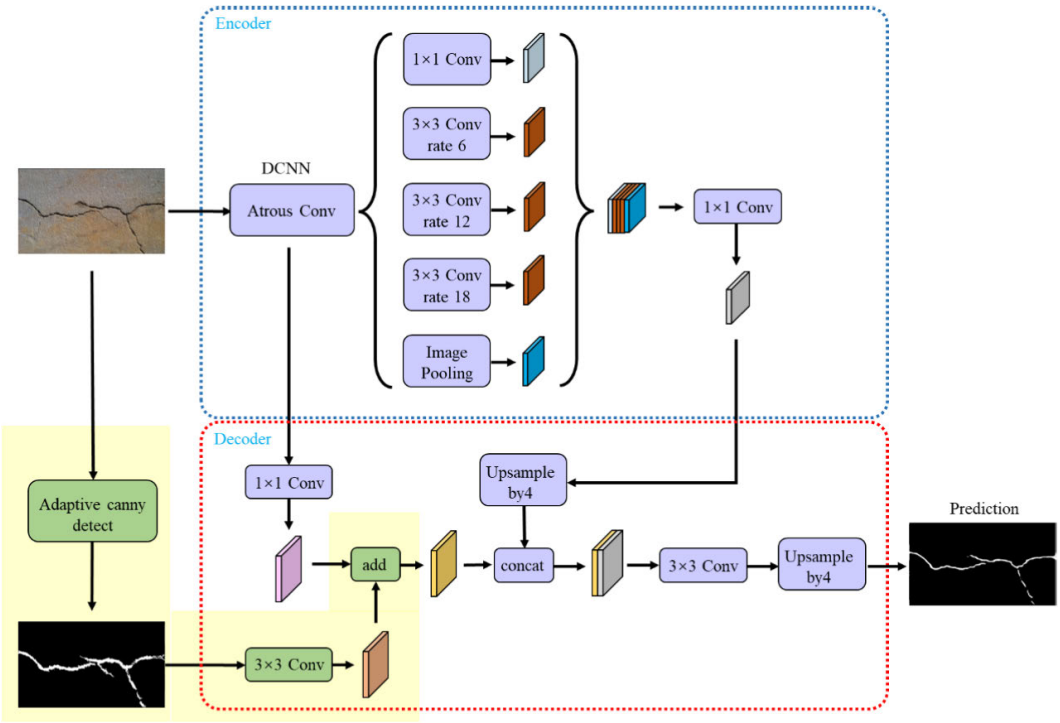

FIGURE 4. Feature fusion structure diagram. The blue box is the encoding part, the red box is the decoding part, and the highlighted area is the innovative method proposed in this paper.

- Backbone is the Xception network,
- The total loss function is the sum of pixel-wise cross entropy loss and auxiliary loss,
- The input image size is 512×512,
- The downsampling factor is 16,
- The optimizer is sgd, the momentum is 0.9, and the weight decay is 0.0001,
- The learning rate drop method is cos,
- Use pre-trained weight files for training

In addition, in the experimental study, intersection over union (IoU), MIoU, Precision, Recall, F1-Score, Pixel Accuracy(PA), and MAE were used to evaluate the performance of the proposed method. The mathematical expressions for these performance metrics are as follows:

$$IoU = \frac{TP}{(TP + FP + FN)} \tag{10}$$

$$MIoU = \frac{(IoU1 + \ldots + IoUn)}{n} \tag{11}$$

$$Recall = \frac{TP}{(TP + FN)} \tag{12}$$

$$Precision = \frac{TP}{(TP + FP)} \tag{13}$$

$$F1 - Score = \frac{(2 \times Precision \times Recall)}{Recall + Precision} \tag{14}$$

$$PA = \frac{(TP + TN)}{(TP + TN + FP + FN)} \tag{15}$$

$$MAE = \frac{1}{N}\sum_{i=1}^{N} \left|y_i - \hat{y}_i\right| \tag{16}$$

where TP, TN, FP, and FN indicate a true positive, a true negative, a false positive, and a false negative, respectively.

This paper selects MIoU and MAE as the final performance indicators to evaluate the algorithm. MIoU measures the intersection and union relationship between predicted and actual regions and then calculates the average. It comprehensively reflects the model performance by considering the similarity and overlap between predicted and actual values. On the other hand, MAE is used to evaluate the model's prediction accuracy by representing the average difference between the predicted and true values. In binary classification semantic segmentation, MAE can assess the model's classification accuracy on a pixel level and directly reflect the difference between the model prediction and actual results. The reason for selecting MIoU and MAE is that they can evaluate the performance of the binary classification semantic segmentation algorithm from different aspects.

B. DATASETS

1) CRACK500

This study used the publicly available complex road cracking dataset CRACK500 to evaluate the performance of the proposed method and compare the performance of other semantic segmentation networks.

Fig. 5 and Table 1 present the CRACK500 dataset and sample sizes used in this study. The dataset is from the main campus of Temple University in the USA, with image sizes of approximately 2000 × 1500 pixels; each road crack image has a pixel-level annotated label, making this dataset the

TABLE 2. CRACK500 details.

Data set	Training set	Test set	Validation set
CRACK500	400	300	50

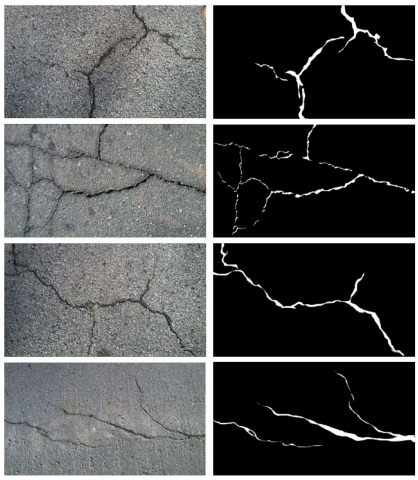

FIGURE 5. Sample images in CRACK500. The left is the original image, and the right is the labeled image.

largest publicly accessible pavement crack dataset with pixel-level annotation.

2) GAPS384

This study evaluates the generalization and limitations of the proposed method based on the model trained on CRACK500 by utilizing the publicly available road crack dataset GAPS384.

TABLE 3. GAPS384 details.

Data set	Training set	Test set	Validation set
GAPS384	180	102	102

Figure 6 and Table 3 illustrate the GAPS384 dataset and sample size used in the experimental study. GAPS384 is a road crack detection dataset maintained by the University of Duisburg-Essen in Germany. The dataset mainly consists of high-definition asphalt concrete road photos. Although the dataset only provides image files without any information about the location and shape of road cracks, Yang et al. [35] cropped and annotated the images with a size of 540 × 440. The dataset is diverse, with various noises such as dark environments, oil stains, and zebra crossing, making it ideal for testing the network's generalization ability.

C. ABLATION EXPERIMENT RESULTS AND ANALYSIS

This paper thoroughly tested and analyzed the adaptive Canny method components proposed. Throughout the experiment, each module used the same parameters and displayed the Canny edge detection image and the image after the morphological operation. Finally, the results of each method were examined on the CRACK500 dataset. The visualization results and values obtained from this study are presented in Figure 7 and Table 4, respectively.

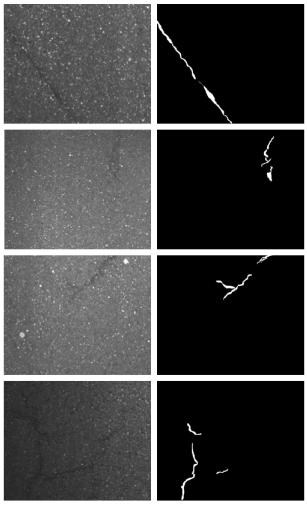

FIGURE 6. Sample images in GAPS384. The left is the original image, and the right is the labeled image.

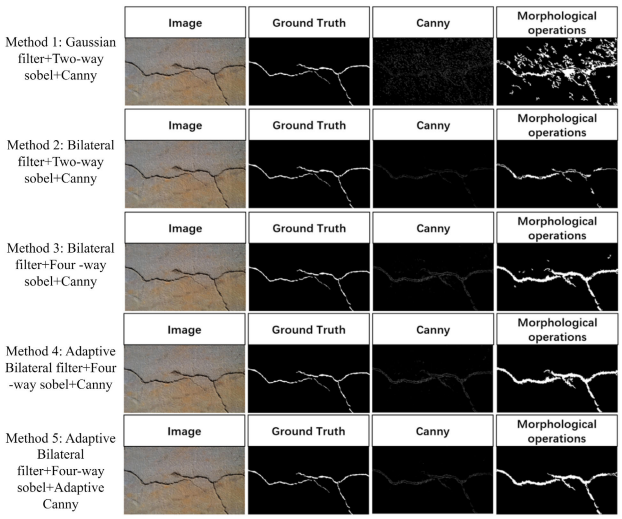

FIGURE 7. Visualization result map of ablation experiment.

TABLE 4. Performance comparison of different models.

Method	Components				CRACK500	
	Bilateral Filter	Four-way sobel	Adaptive Bilat-eral Filter	Adaptive Canny	MIoU%	MAE
Method1	×	×	×	×	19.39	26.44
Method2	✓	×	×	×	37.61	9.76
Method3	✓	✓	×	×	52.33	9.51
Method4	✓	✓	✓	×	57.17	9.02
Method5	✓	✓	✓	✓	65.28	6.24

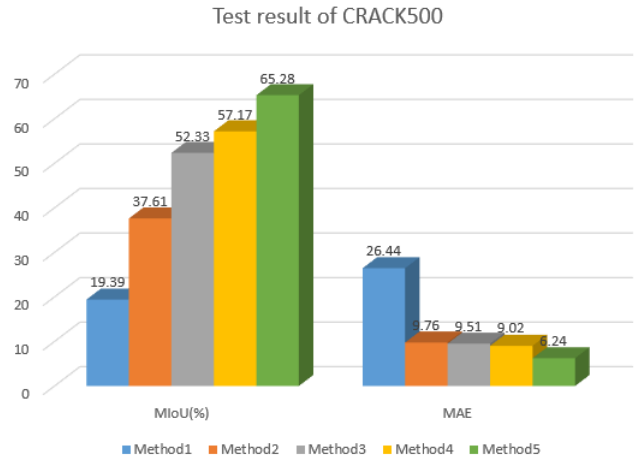

FIGURE 8. Histogram of the test results of the ablation experiment on CRACK500.

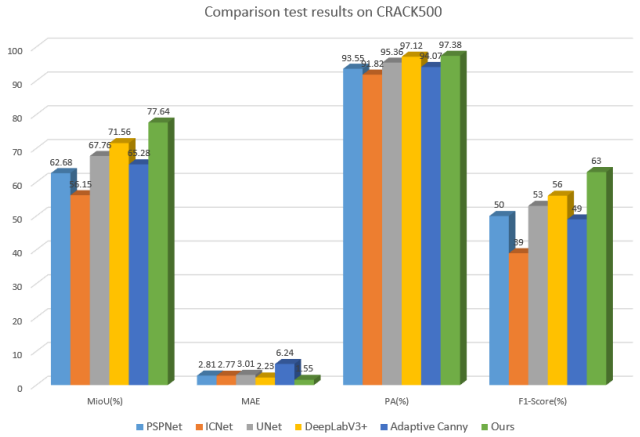

FIGURE 9. Histogram of comparative test results of each algorithm in CRACK500.

We compared and analyzed different methods' road crack detection capabilities based on the detection results in Table 4 and the performance comparison in Figure 8. Method 1 exhibited poor anti-interference and edge feature extraction, as shown in Figure 8. The Canny image generated by it contained a significant amount of noise, and the road crack edge was not clear enough. Additionally, after the morphological operation, the resulting map differed from the real value. Method 2 improved on method 1 by using bilateral filtering instead of Gaussian filtering, which significantly enhanced the effect of denoising and edge preservation. Despite eliminating many environmental noises, it failed to extract all the road crack edge details. Method 3 enriched fracture edge information by introducing the four-way Sobel operator based on method 2, but also mistakenly extracted some environmental noise. Method 4 further reduced interference by adding adaptive bilateral filtering based on method 3 and continuously adjusting the filtering parameters through the feedback of the resulting map. However, the detected road crack edge was still not accurate enough. Method 5 adapted Canny detection based on method 4, which enabled the algorithm to select the Canny threshold adaptively according to the image gradient information, achieving better detection results.

The evaluation indicators used to measure the detection results of each method on the CRACK500 dataset were the MIoU value and the MAE value, as shown in Table 4. The MIoU value represents the overlap between the detected and actual areas. In contrast, the MAE value represents the difference between the average pixel of the detected image and the actual image. From Table 4, we can see the algorithm modules of each method and their detection results MIoU and MAE tested on the CRACK500 dataset. Figure 8 visually demonstrates the performance improvement of each method after adding different algorithm modules. From method 1 to method 5, the MIoU value gradually increased, and the MAE value gradually decreased. Method 5 improved MIoU by approximately 45% and MAE by approximately 20 compared to method 1. Therefore, this experiment proves that each new module can improve the road crack detection ability of the algorithm, and each algorithm module proposed in this paper is reliable and necessary.

D. COMPARATIVE EXPERIMENT RESULTS AND ANALYSIS

The CRACK500 dataset includes small-sized surface defects that resemble the background and images taken under varying lighting conditions, making it a complex dataset for surface defect detection.

Fig. 10 shows the numerical IoU and visual estimates of some surface examples and methods for the CRACK500 dataset. Fig. 10 displays ground fracture scenarios for different materials, with examples of larger background disturbances, as in rows 1, 6, and 7 and images with minor surface defects, as in row 5. As ICNet is a lightweight network, the IoU values for ICNet detection, as shown in Fig. 10, were not high compared to those for other methods, especially for the images in row 5, which contain more road crack details. The DeepLabV3+ network detected the various road crack example images in Fig. 10 better than the other methods; however, tiny details are lost in the example images in rows 1, 2, and 7. Adaptive Canny detected road cracks better in the example images; however, noise interference was observed in the road crack images in rows 1, 6, and 7, which had high background interference. There was excessive detail loss in the example image in row 5. In addition, the adaptive Canny method suffered from insufficient smoothing of the detected road crack edges.

The method of adaptive Canny fused with DeepLabV3+ proposed in this study showed higher IoU values in road crack example map detection than other methods, as shown in Fig. 10. The detection in rows 1, 6, and 7 where the background environment is more disturbed, is satisfactory, and the road crack body can be detected by removing the background interference. Detection is also improved over other methods for row 5, which contains a large area of road cracks with

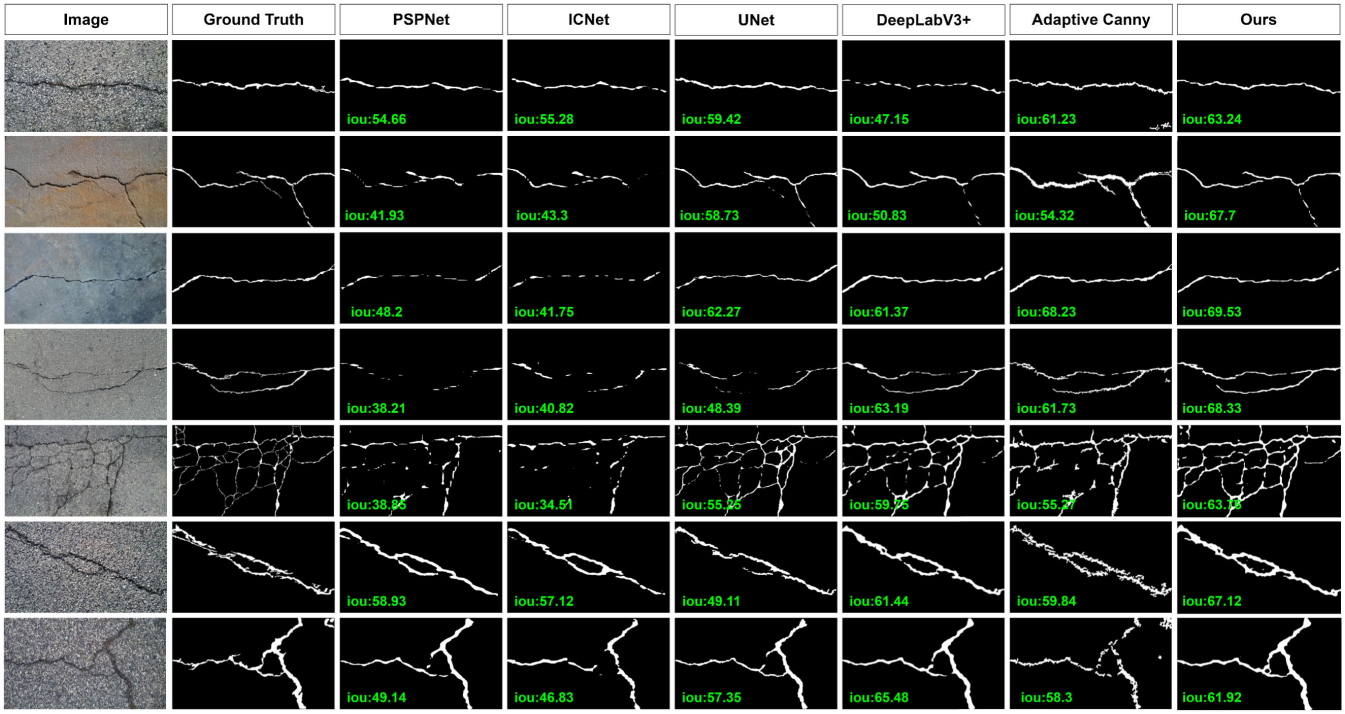

FIGURE 10. Visualisation results from the CRACK500 dataset.

TABLE 5. Test results (%) of the proposed method and other test results on CRACK500.

Model	MIoU(%)	MAE	PA(%)	F1-Score(%)
PSPNet	62.68	2.81	93.55	50
ICNet	56.15	2.77	91.82	39
UNet	67.76	3.01	95.36	53
DeepLabV3+	71.56	2.23	97.12	56
OpenCV	65.28	6.24	94.07	49
Ours	**77.64**	**1.55**	**97.38**	**63**

more details. The IoU scores of the proposed method are higher than those of the DeepLabV3+ and adaptive Canny methods in the example map detection and higher than those of the other networks.

The proposed method in this paper was tested three times with other semantic segmentation networks on the CRACK500 dataset, and the average values of the evaluation indicators are presented in Table 5. Figure 9 visually illustrates the performance of each algorithm, allowing for a more intuitive comparison of their pros and cons in detecting road cracks.

As the MIoU and Mean MAE can measure the performance of binary classification semantic segmentation algorithms in different aspects, this paper selects them as the final evaluation indices. The fusion method proposed in this paper obtained the highest score, with an MIoU of 77.64% and an MAE of 1.55, by integrating the adaptive Canny and DeepLabV3+ methods. The adaptive Canny method accurately detects small road cracks and uses this as a prior on the DeepLabV3+ network, while the DeepLabV3+ method smoothes the road crack edges and makes up for the parts not detected by Canny. The DeepLabV3+ method scored the second highest, with an MIoU of 77.64% and an MAE of 2.23. Its success can be attributed to the introduction of the resolution, which allows for control of the feature extraction of the encoder, balance of accuracy and time consumption through atrous convolution, and increase of the receptive field without losing information, resulting in each convolution output containing an extensive range of information. Adaptive Canny's MIoU score ranked fourth among all methods, and it performed poorly on the MAE index, mainly due to the non-crack area pixels that improved the MAE score.

Therefore, the information provided in Figure 9 and Table 5 suggests that the method proposed in this paper has better detection results on the CRACK500 dataset and improved detection results compared to the adaptive Canny, DeepLabv3+, PSPNet, ICNet, and Unet networks. It is difficult to cope with the complex and changing environment by using only the adaptive Canny method because it is easily disturbed by noise when detecting in a noisy environment, resulting in false detection in the final detection result map. When only DeepLabV3+ is used for detection, some details will be lost when fusing and splicing the high-feature and low-feature layers, unfavorable to the final detection result. The method proposed herein takes advantage of each and the combined detection. The combined detection method can overcome the interference of complex environmental noise. It can also be used to accurately detect complex cracks with

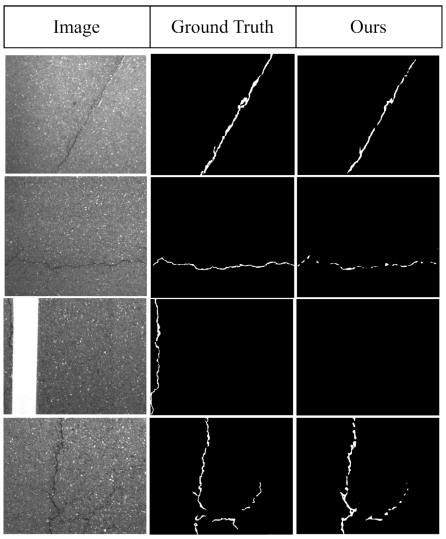

FIGURE 11. The trained model is tested in GAPS384.

TABLE 6. The trained model is tested in GAPS384.

Model	MIoU	MAE	PA(%)	F1-Score(%)
Ours	51.29	3.58	88.52	46

many details, such that the final results are better than those of other comparison methods.

We evaluated the generalization capability and limitations of the proposed method by applying it to a different dataset, namely GAPS384, after training it on the CRACK500 dataset. The results of our experiments are presented in Figure 11 and Table 6, which include both visual results and numerical values.

Based on the results presented in Figure 11 and Table 6, the proposed method in this study demonstrates a certain degree of generalization ability on different datasets. It can reliably detect road cracks even after scene changes. However, this method has some limitations, specifically regarding the road crack images in the GAPS384 dataset, which originate from asphalt pavement. Due to the small difference between the road crack and the road surface, the presence of low light, and interference from road signs, such as zebra crossings, this method is unsuitable for detecting road cracks in this case, resulting in a weak detection ability. The detection results are displayed in lines 2 and 3 of Figure 11. Furthermore, the information extracted from the road cracks by the proposed method is insufficient, and the road cracks cannot be successfully detected.

V. CONCLUSION

This study proposed a feature fusion-based adaptive Canny and semantic segmentation network for road crack detection. Based on improved Canny edge detection, the method fuses the Canny crack detection map with the low feature layer in the DeepLabV3+ detection network after extracting information from the convolutional feature extraction module. Finally, it splices it with a high feature layer after the resulting map is obtained. The experimental results on the CRACK500 dataset demonstrate the effectiveness of the proposed fusion method, with an MIoU of 77.64 and an MAE of 1.55, outperforming other semantic segmentation detection algorithms. This method can reliably eliminate interference and detect a more comprehensive range of fracture details. However, it still has limitations, with detection performance deteriorating in dark environments or when interfering objects are present. Thus, future research is necessary to optimize the algorithm further and enhance its detection accuracy in such scenarios while improving its generalization ability.

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